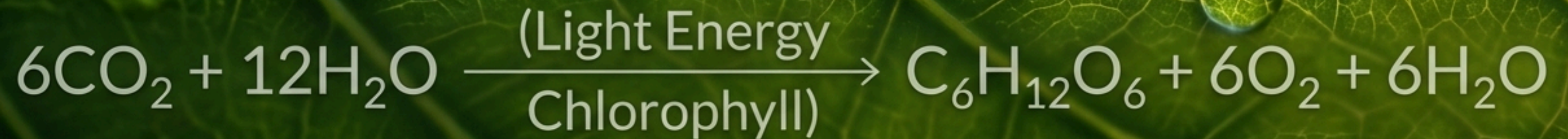


# Photosynthesis: The Engine of Life

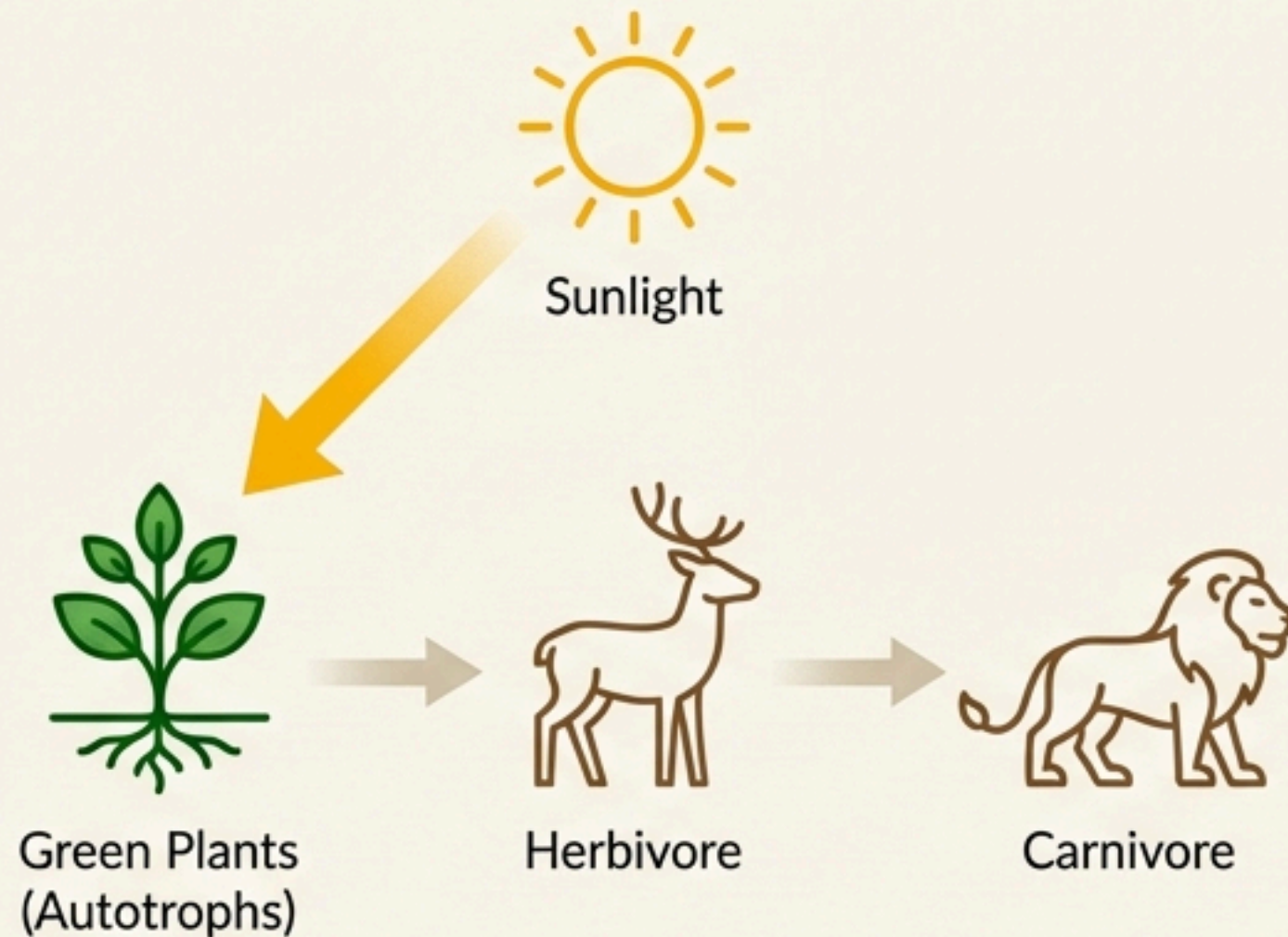
A Visual Journey into How Plants Create Food from Sunlight



# The Most Important Process on Earth

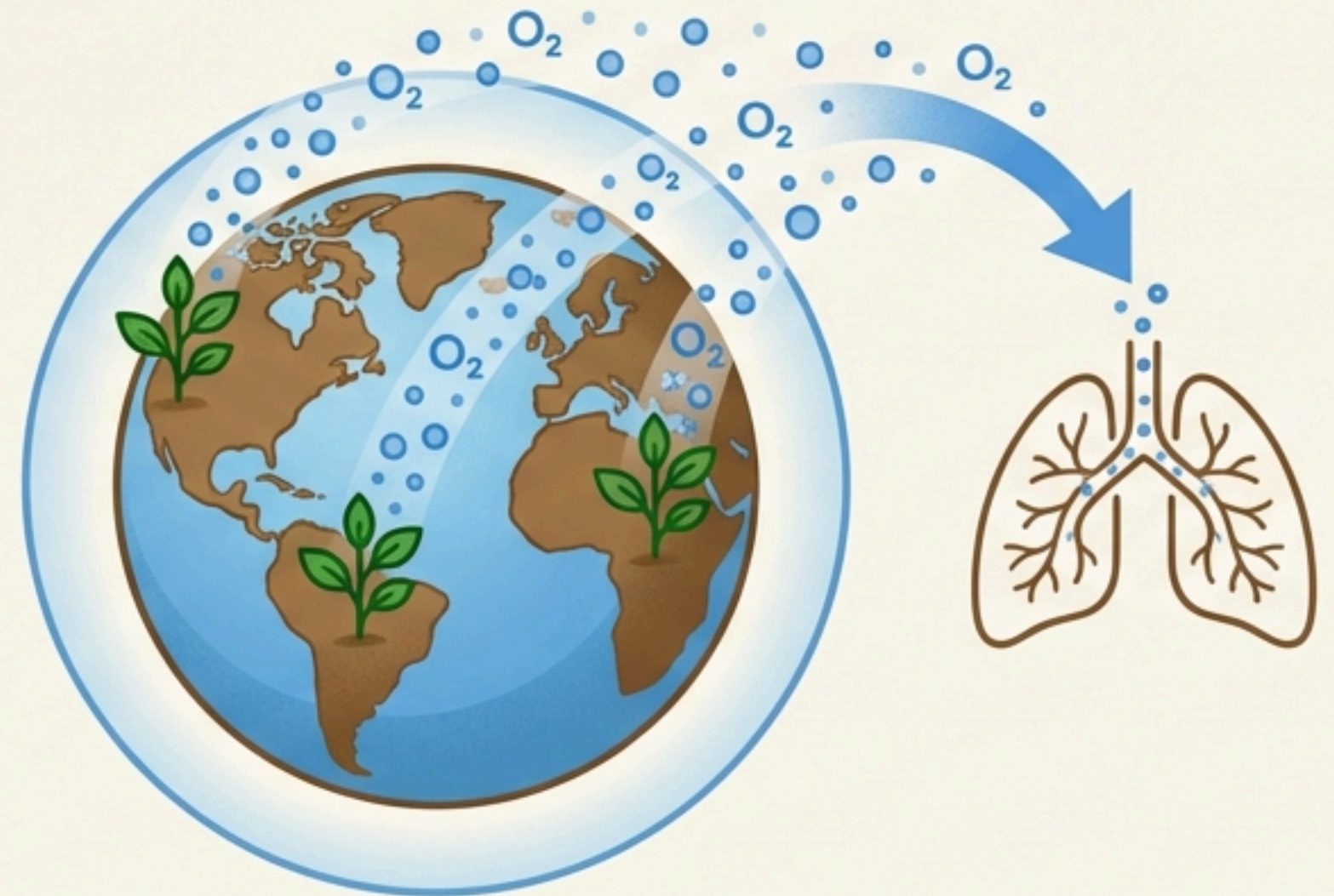
## Primary Source of All Food on Earth

Green plants are autotrophs, synthesizing the food they need. All other organisms, including humans, are heterotrophs that depend on them. Ultimately, all living forms on Earth depend on sunlight for energy, captured through photosynthesis.



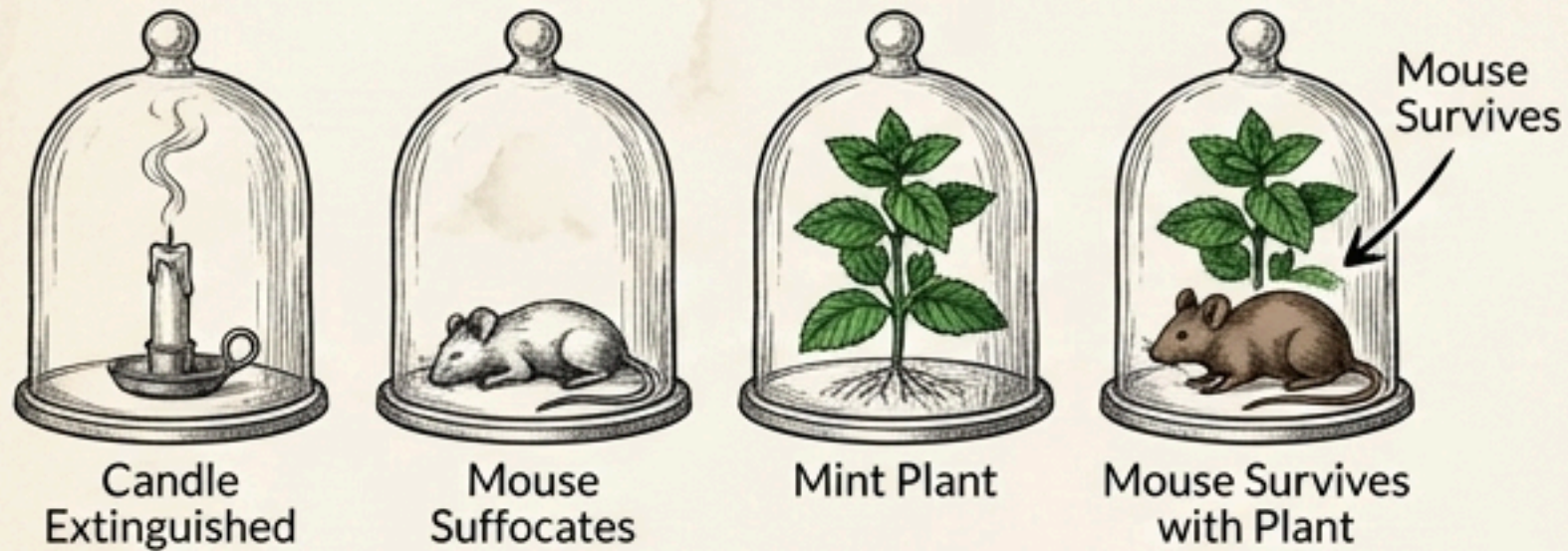
## Responsible for the Oxygen We Breathe

Photosynthesis is responsible for the release of oxygen into the atmosphere by green plants, a crucial element for respiration in most living organisms.



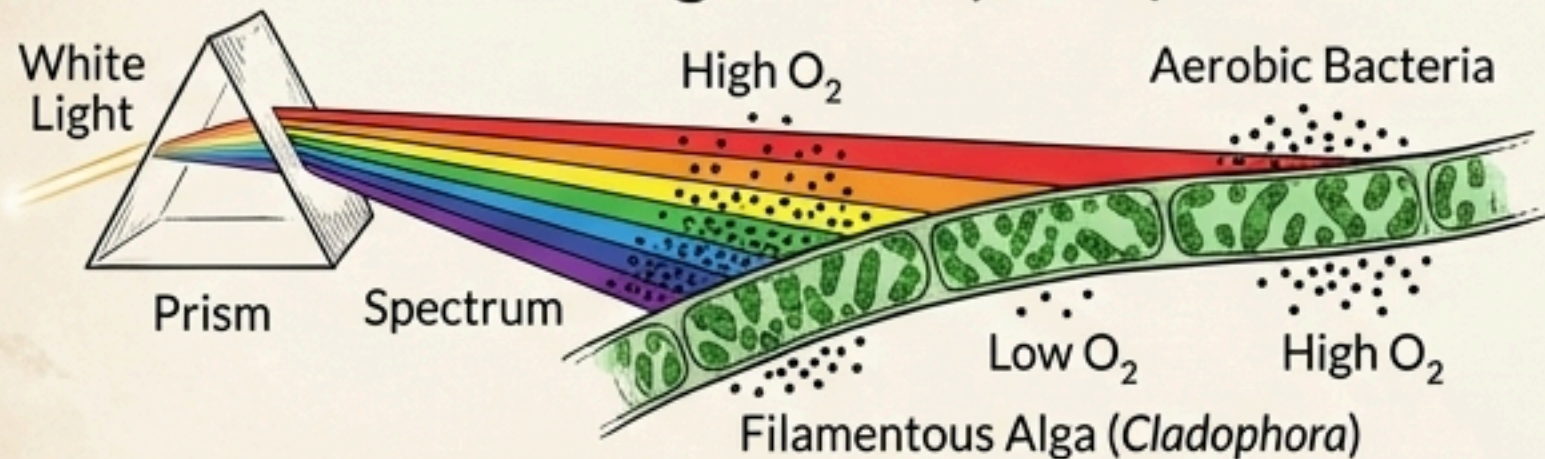
# Piecing Together the Puzzle: A History of Discovery

Joseph Priestley (1770s)



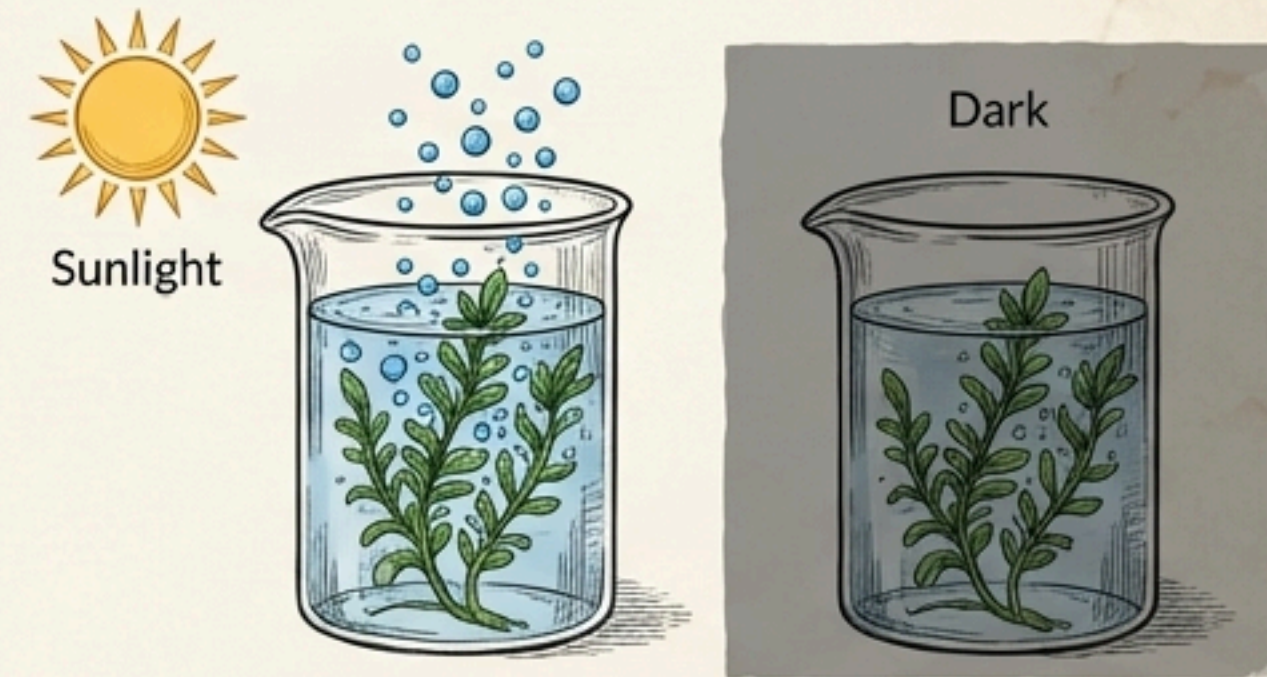
*Plants restore to the air whatever breathing animals and burning candles remove.*

T.W. Engelmann (1843)



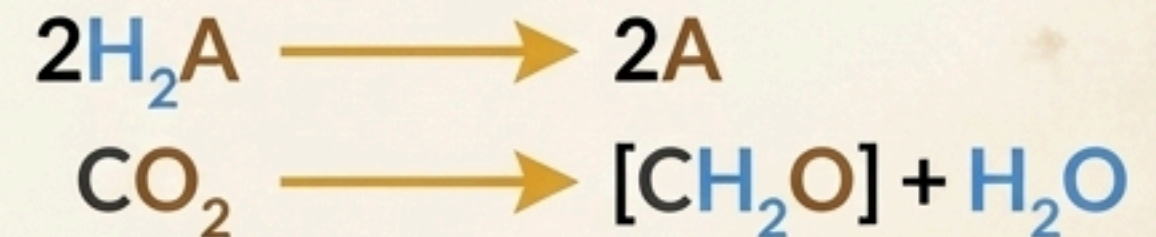
*A first action spectrum was described, showing photosynthesis is highest where chlorophylls absorb light.*

Jan Ingenhousz (1779)



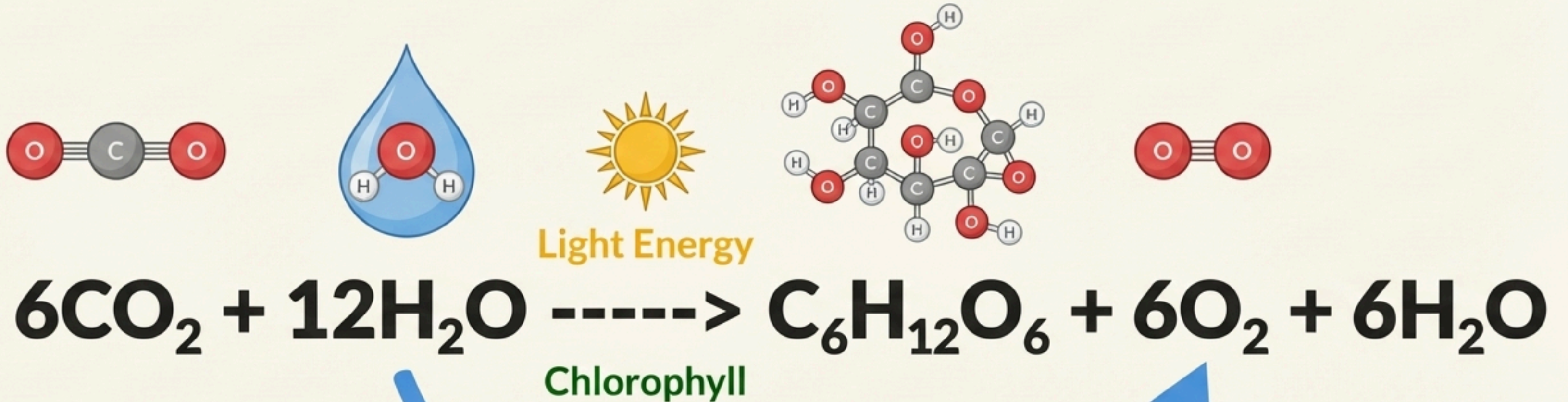
*Sunlight is essential for the plant process that purifies the air.*

Cornelius van Niel (1897–1985)



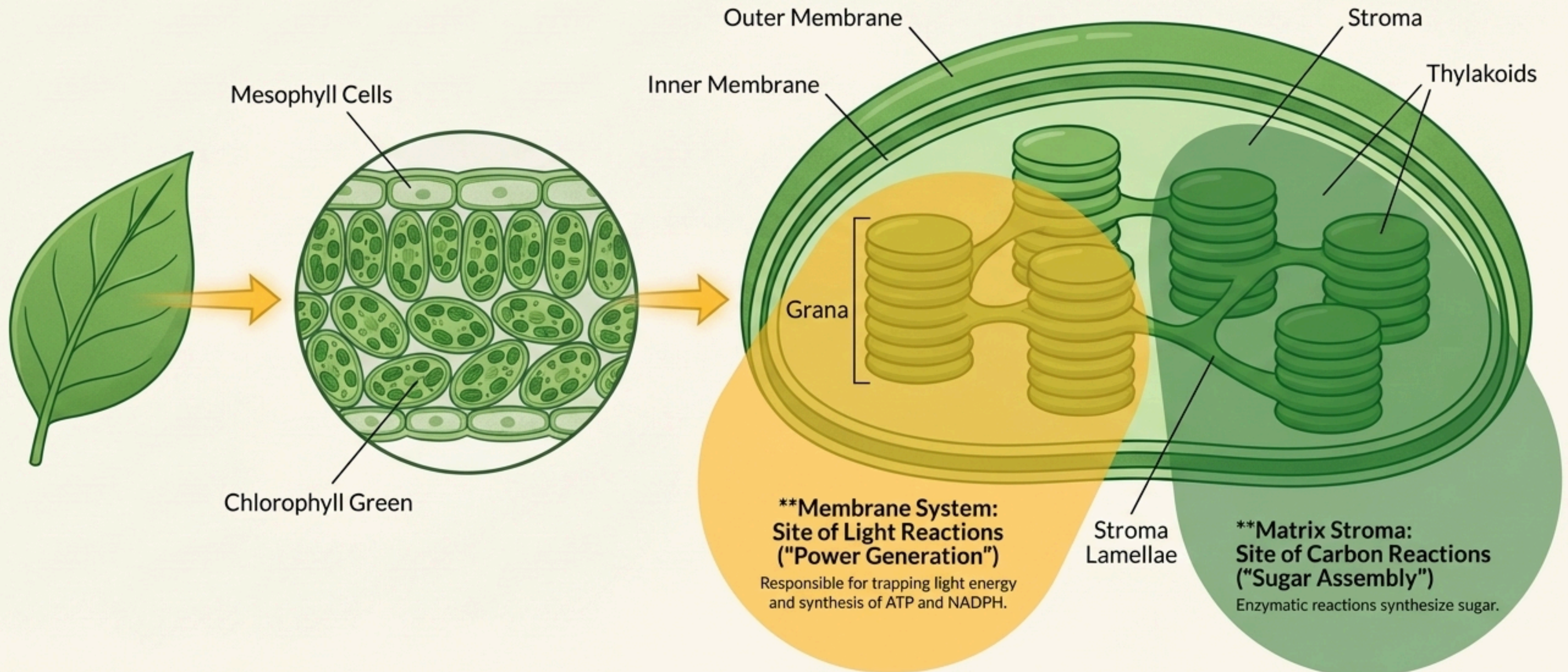
*The O<sub>2</sub> evolved by the green plant comes from H<sub>2</sub>O, not from carbon dioxide.*

# The Chemical Blueprint of Life



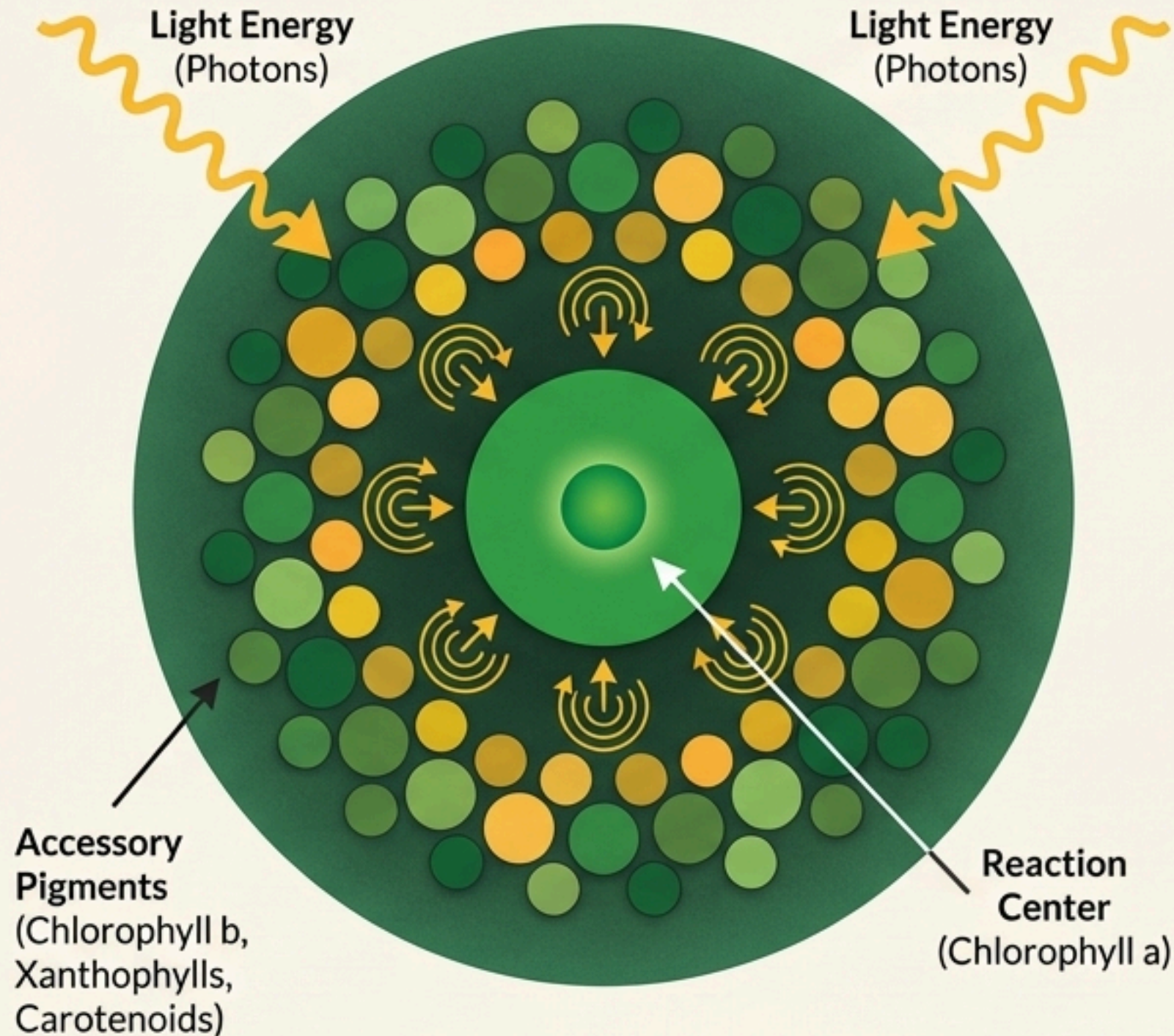
Van Niel's hypothesis was proven using radioisotopic techniques: Oxygen released comes from water.

# The Photosynthesis Powerhouse: The Chloroplast

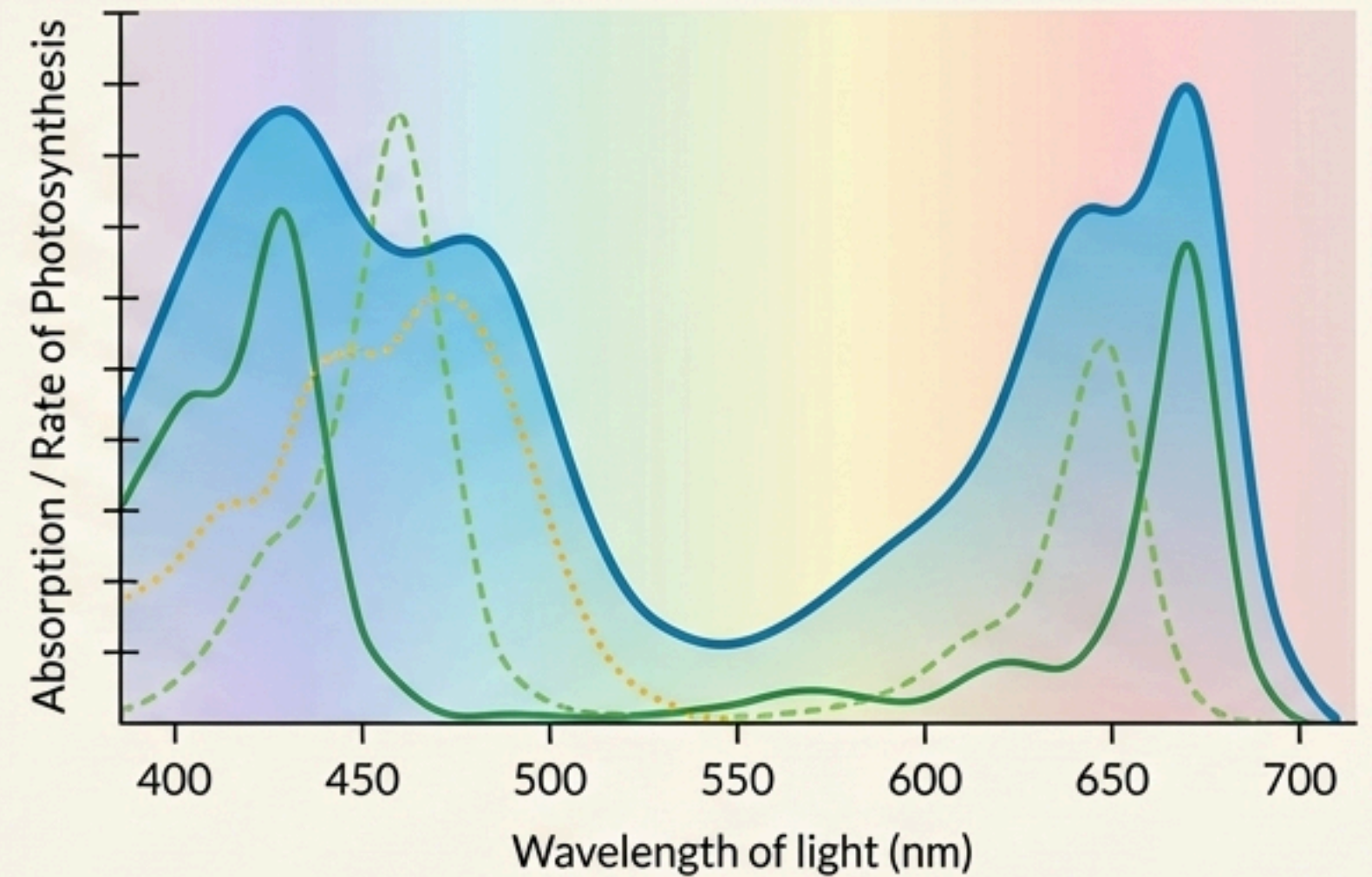


# Act I: Harvesting Light Energy

## The Machinery

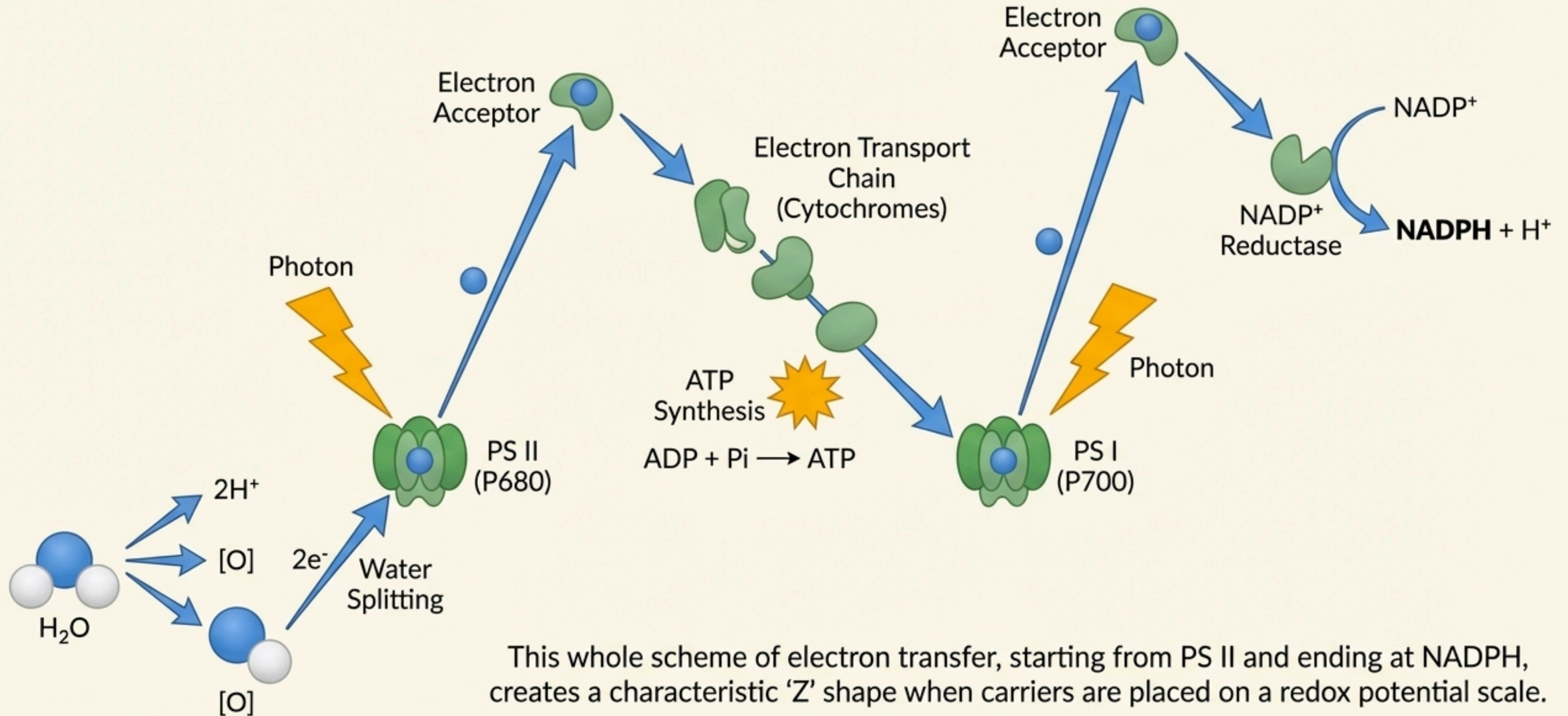


## The Performance



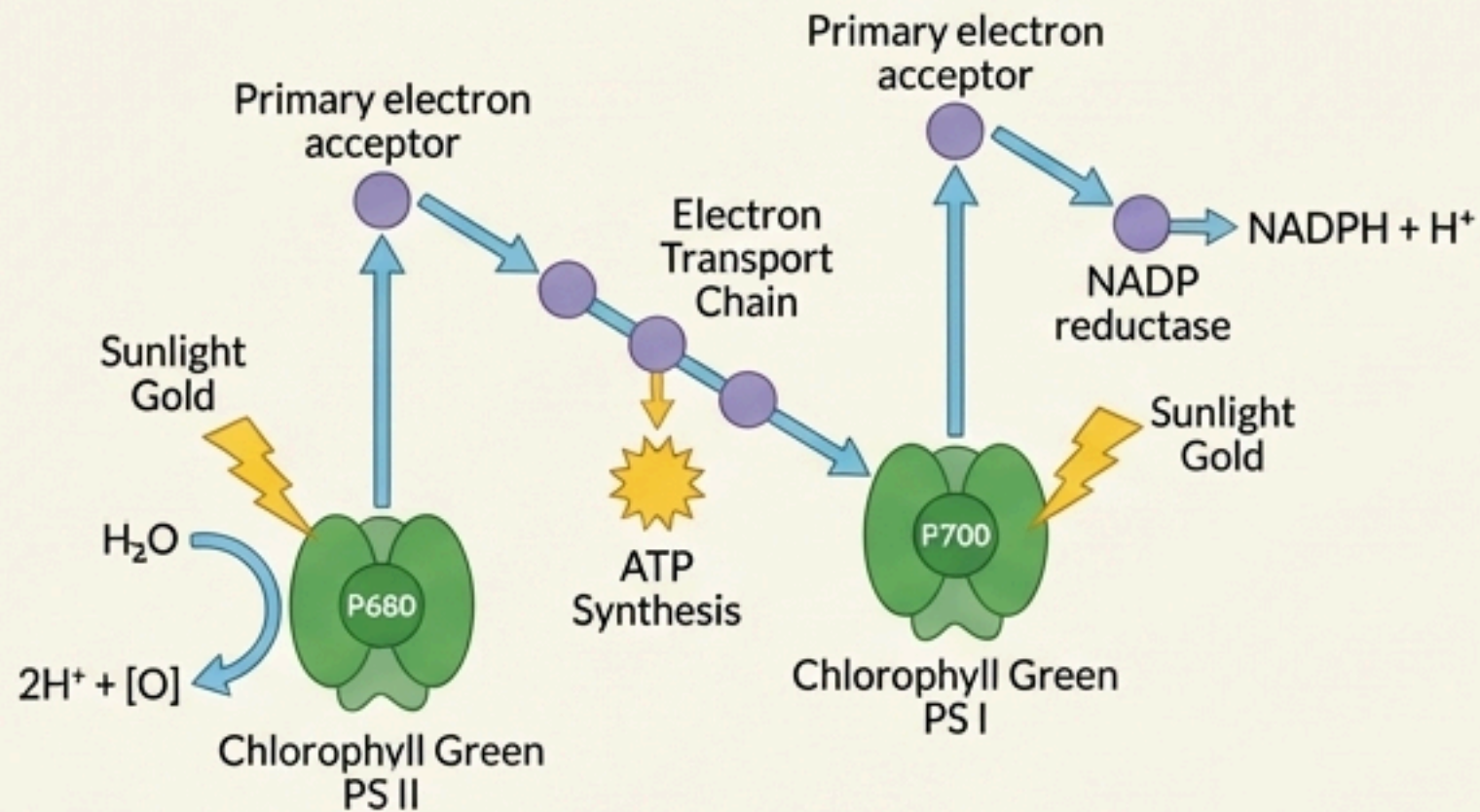
Chlorophyll a is the chief pigment. Accessory pigments broaden the spectrum of usable light and also protect Chlorophyll a from photo-oxidation.

# The Z-Scheme of Non-Cyclic Electron Transport



# Two Paths for Power Generation: Cyclic vs. Non-Cyclic

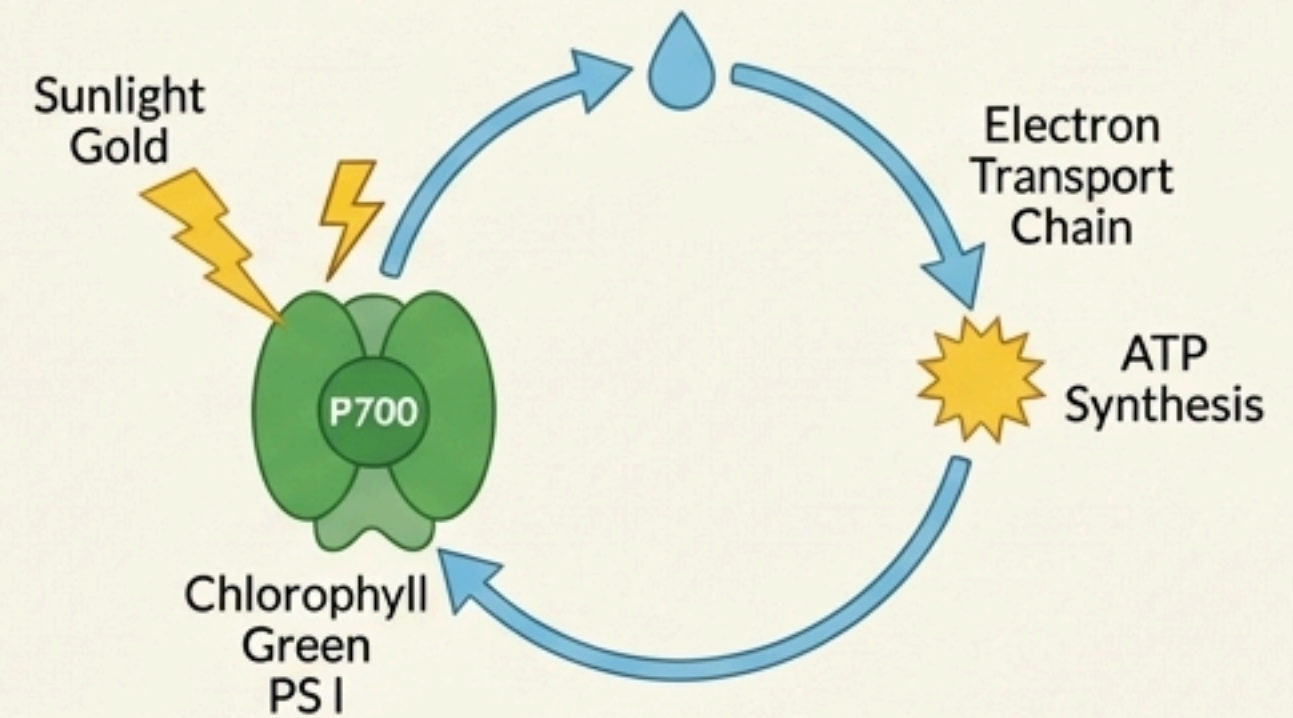
## Non-Cyclic Photophosphorylation



**How it works:** Two photosystems (PS II and PS I) work in series, connected by an electron transport chain.

**Products:** ATP  + NADPH  +  $O_2$  

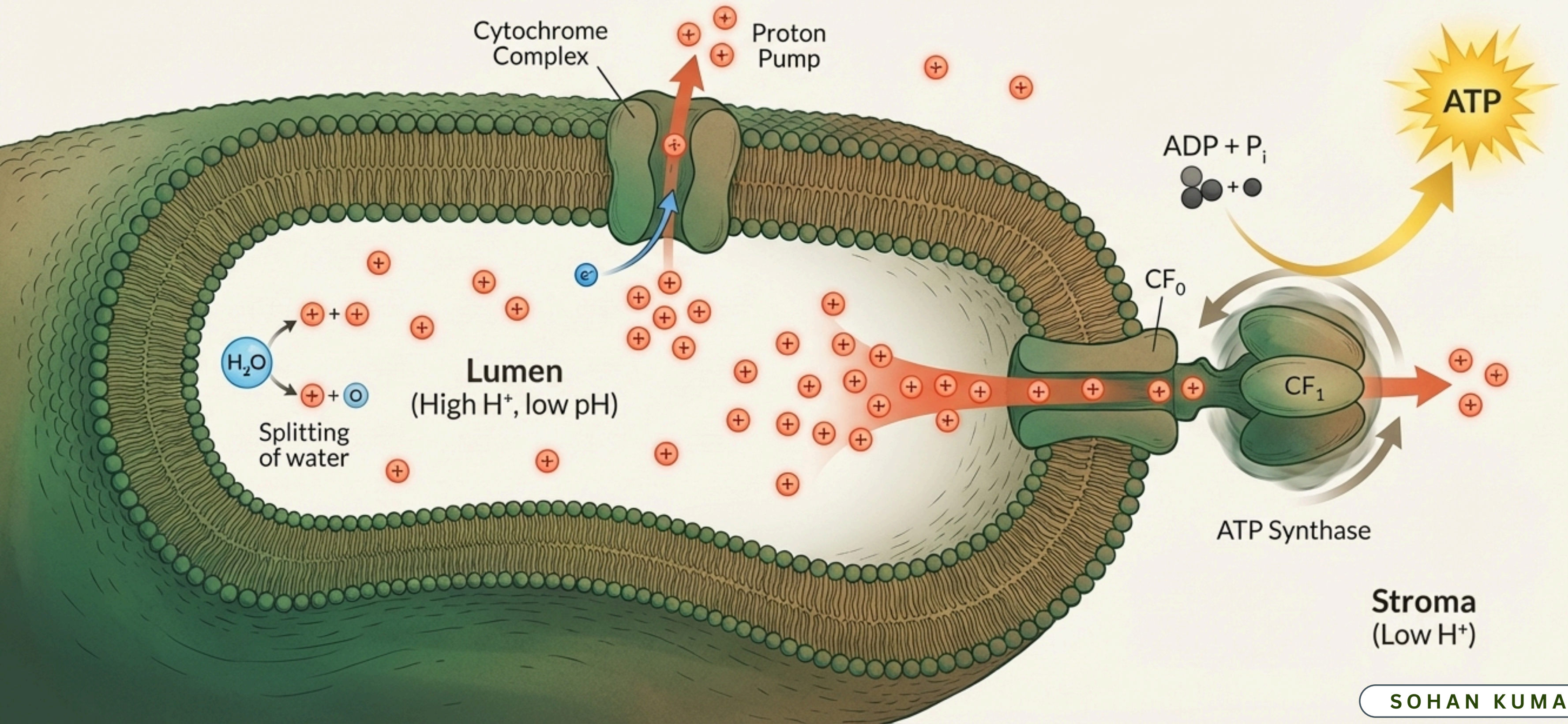
## Cyclic Photophosphorylation



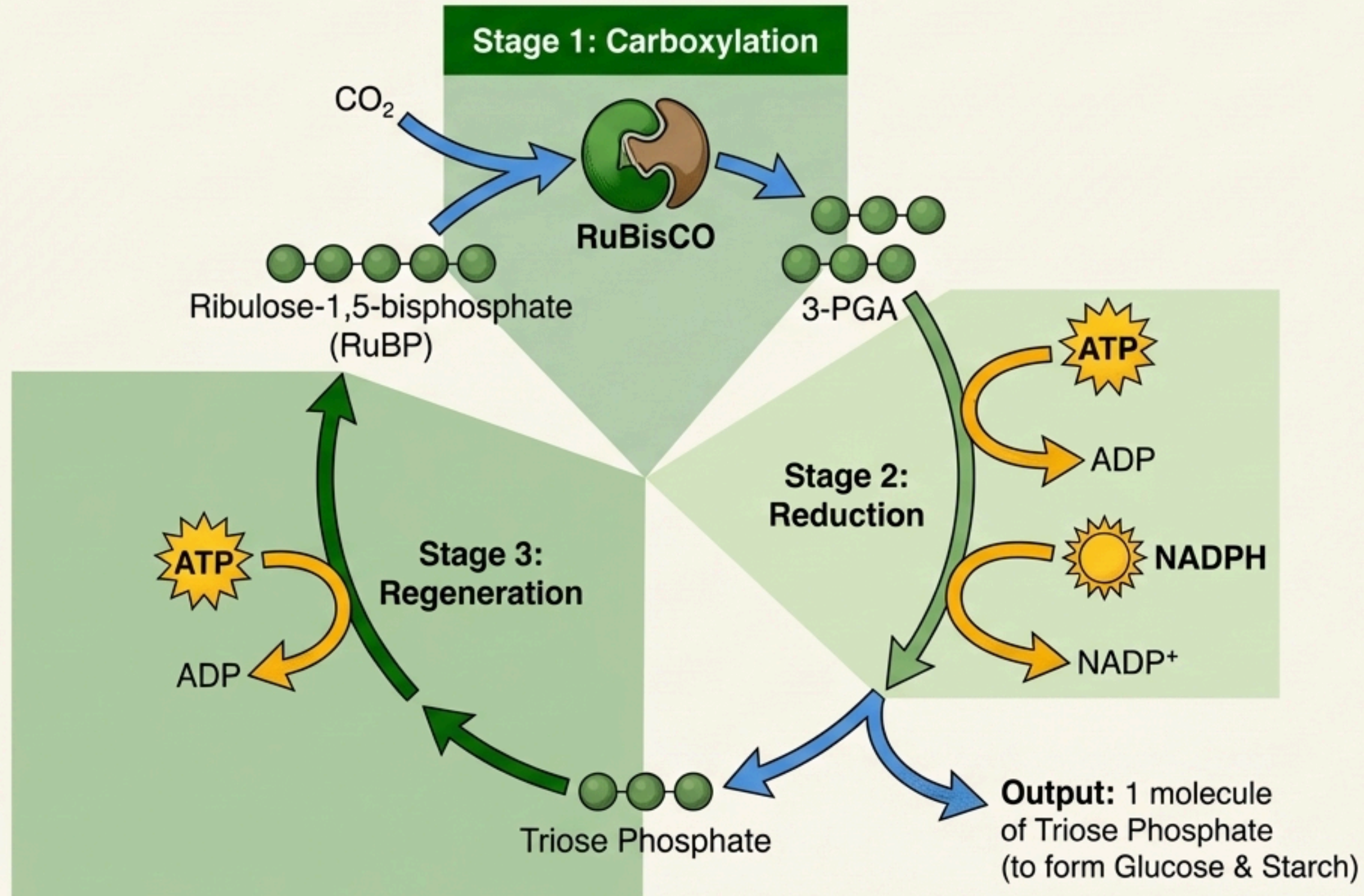
**How it works:** Only PS I is functional. Occurs when light of wavelengths beyond 680 nm is available or in stroma lamellae (which lack PS II and NADP reductase).

**Products:** ATP only   
(No NADPH or  $O_2$  produced).

# Chemiosmosis: The ATP Synthase Engine



# Act II: The Calvin Cycle - Building with Carbon



# The Recipe for One Glucose Molecule

INPUTS

**6x**

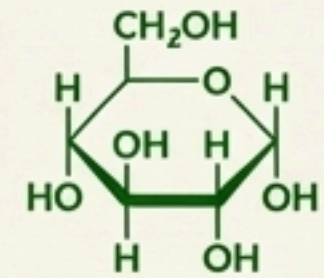


Carbon Dioxide

**To make 1 molecule of Glucose ( $C_6H_{12}O_6$ ), the cycle must turn 6 times.**

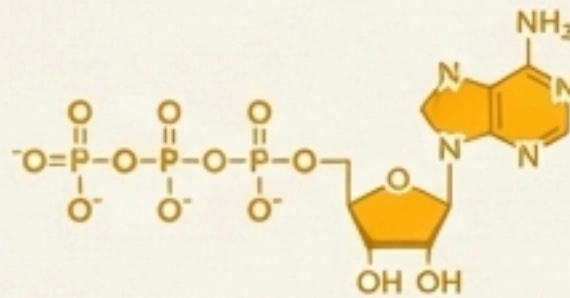
OUTPUTS

**1x**



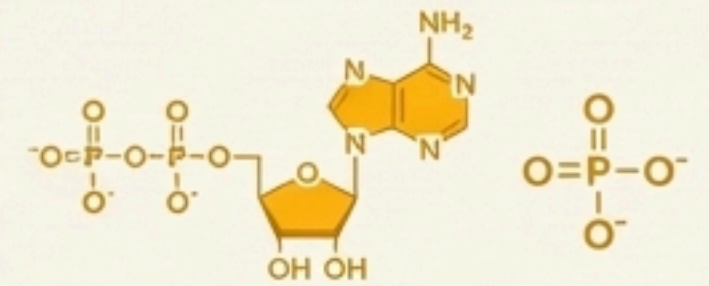
Glucose

**18x**



ATP

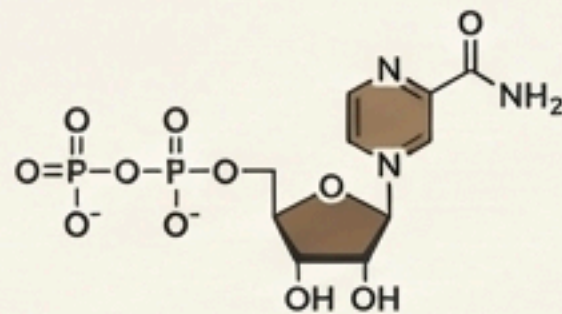
**18x**



ADP

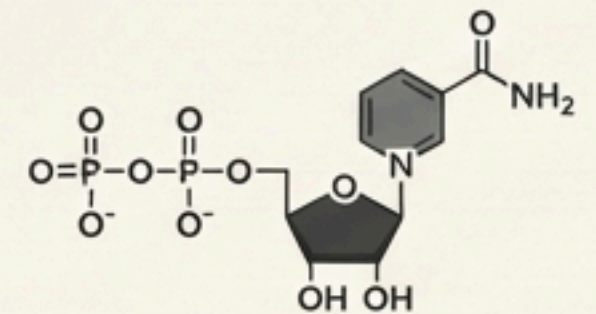
Pi

**12x**



NADPH

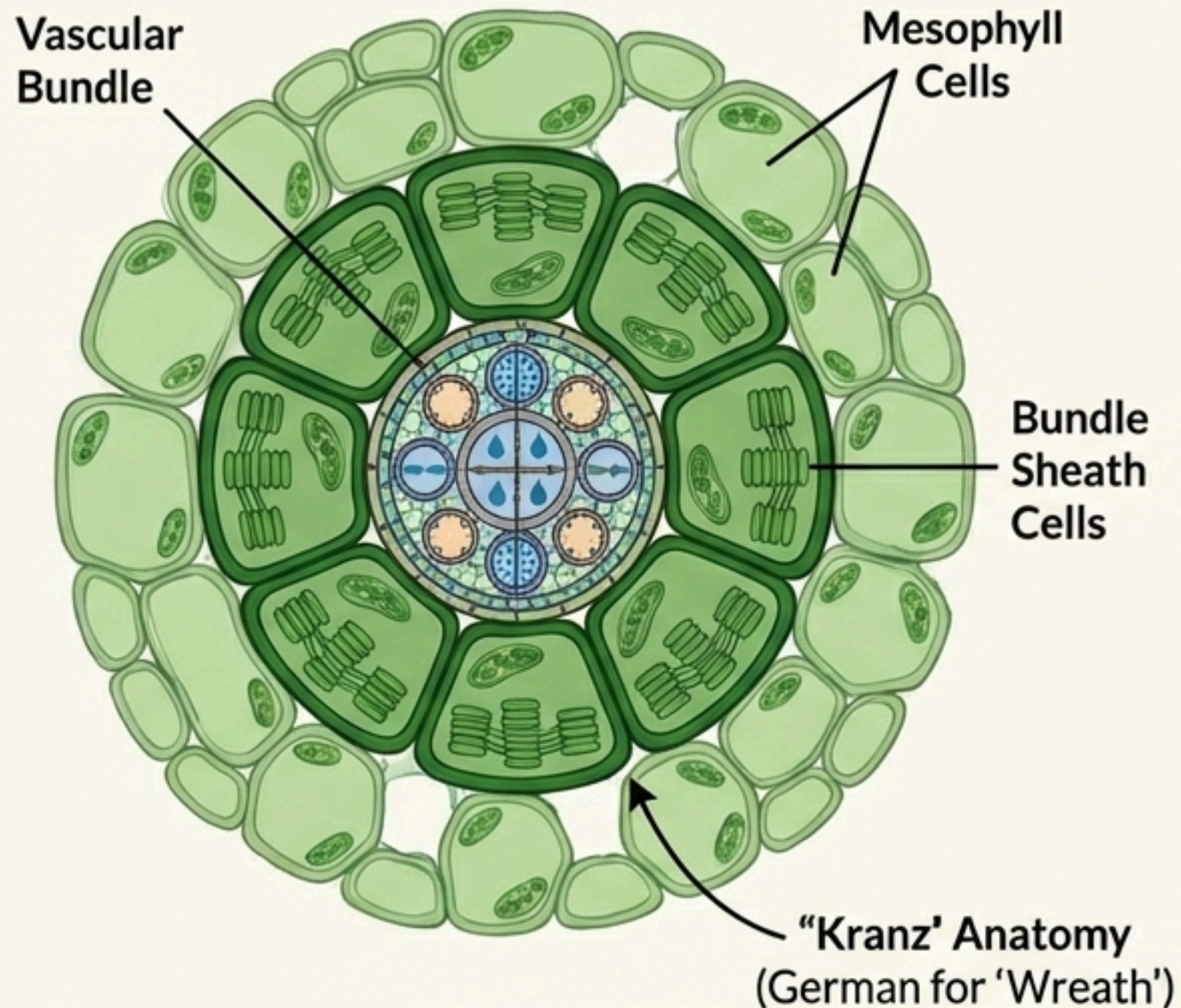
**12x**



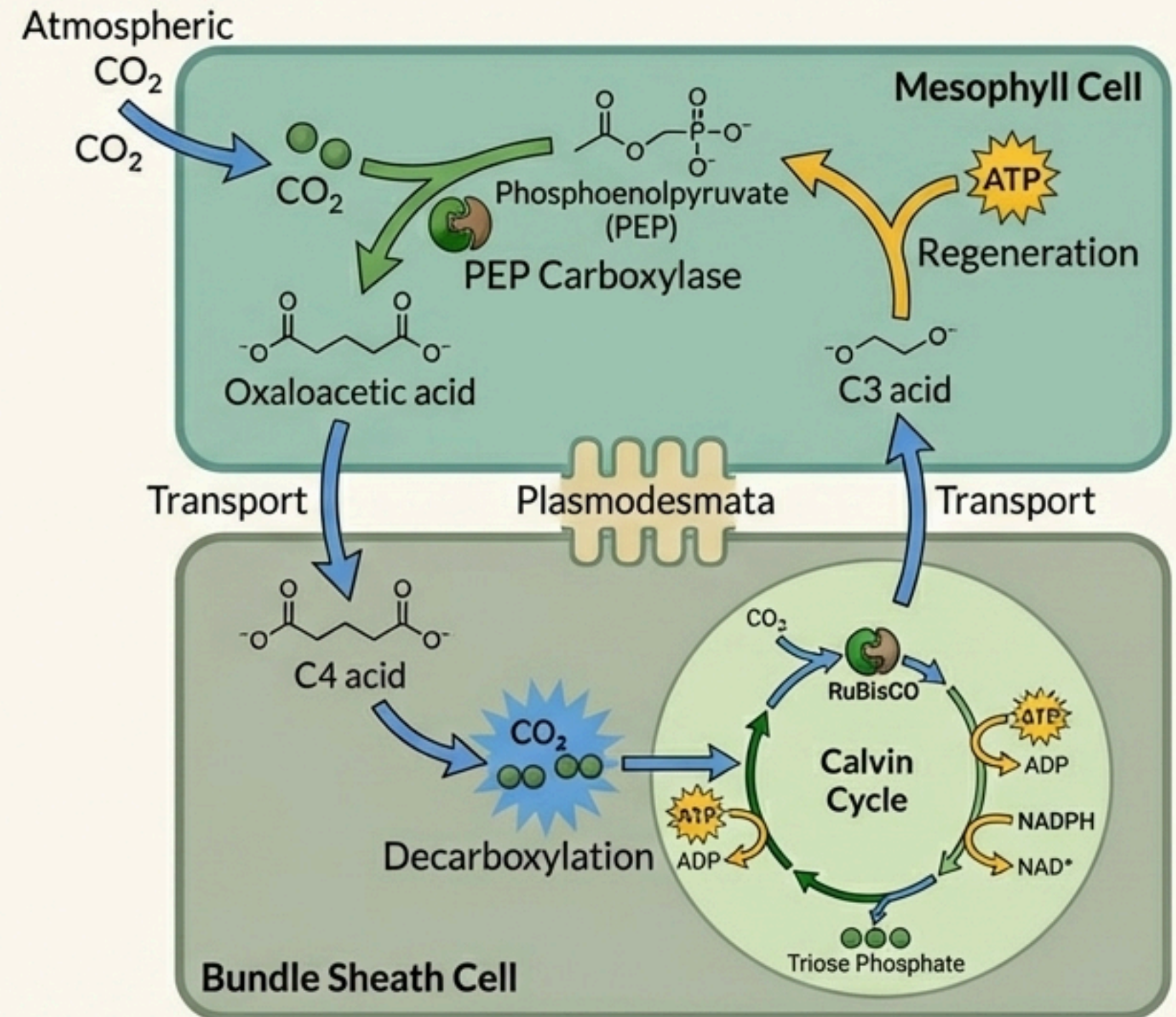
NADP<sup>+</sup>

# The C4 Solution: A High-Efficiency Upgrade for Hot, Dry Climates

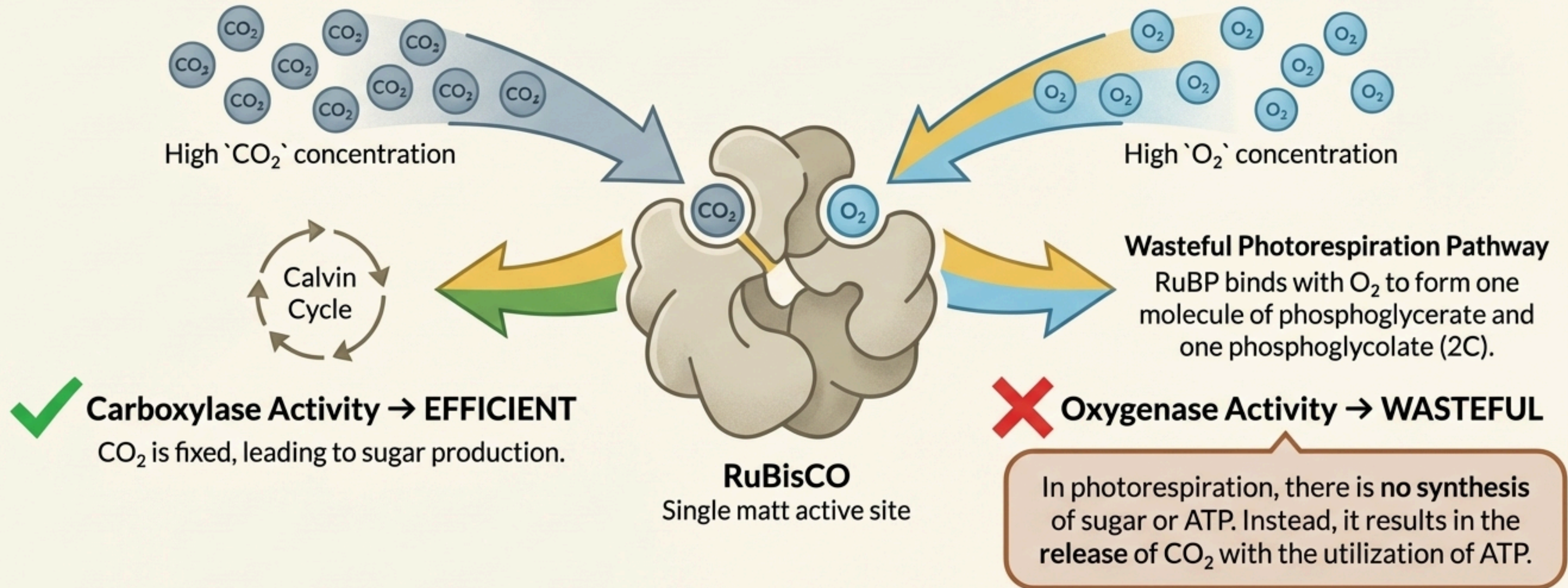
## Kranz Anatomy



## The Hatch-Slack Pathway



# Photorespiration: The Wasteful Problem C4 Solves



C4 plants solve this by using the Hatch-Slack pathway to pump CO<sub>2</sub> into the bundle sheath cells, increasing local CO<sub>2</sub> concentration and minimizing oxygenase activity.

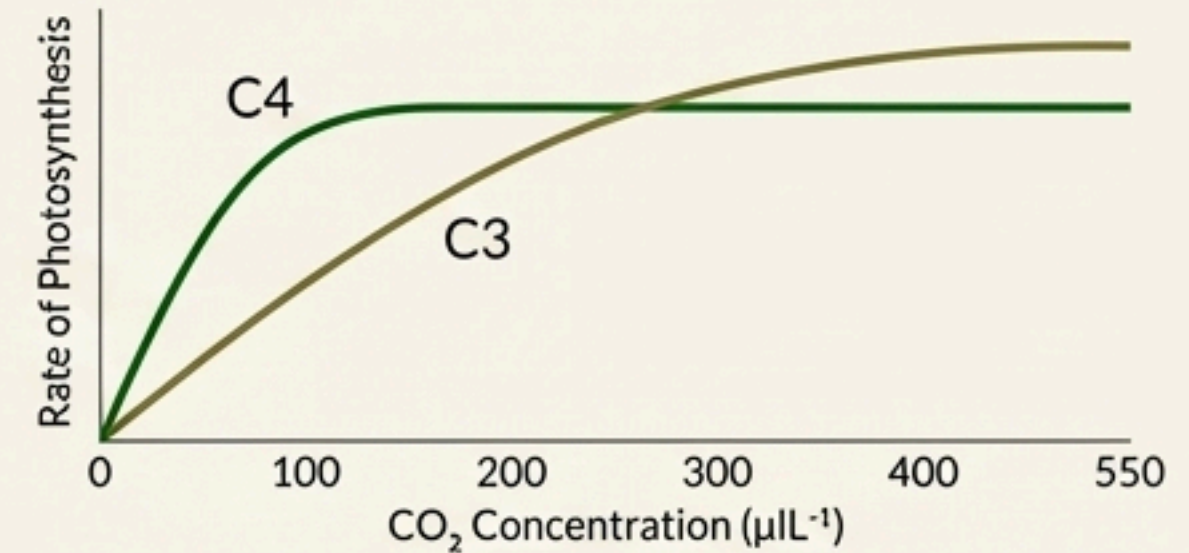
# Factory Output: Limiting Factors

## Light Intensity



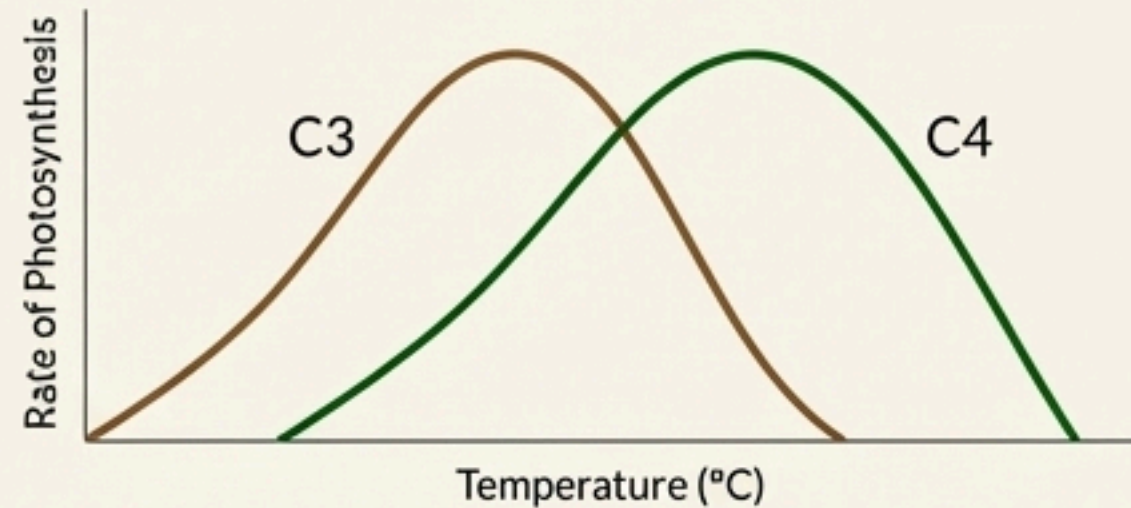
Light saturation occurs at 10% of full sunlight. Above this point, other factors become limiting.

## CO<sub>2</sub> Concentration



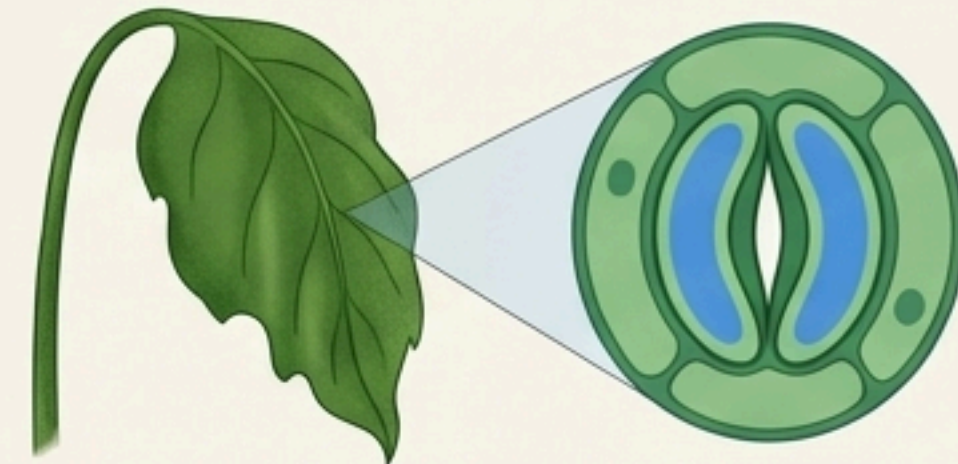
CO<sub>2</sub> is the major limiting factor for C<sub>3</sub> plants at current atmospheric levels.

## Temperature



Dark reactions are enzymatic and temperature-sensitive. C<sub>4</sub> plants have a higher temperature optimum than C<sub>3</sub> plants.

## Water



Water stress affects photosynthesis indirectly by causing stomata to close, which reduces CO<sub>2</sub> availability.

# Photosynthesis at a Glance: The Complete Journey

